Ngan Nguyen

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Education

VNU University of Engineering and Technology (VNU-UET) *Hanoi, Vietnam*
Bachelor of Information Technology *2021 – 2025*
Vietnam National University, Hanoi *GPA: 3.67/4.0*

Research Interests

My current research focuses on **efficient generative modeling**, particularly **one-step diffusion distillation**, **controllable image generation**, and **pixel- and latent-space diffusion**. I am also interested in exploring **vision-language models (VLMs)**, **agentic AI**, **video generation**, and **world models** in future research.

Publications

Conference Publications ∗ denotes equal contribution

4. **FlashControl: One-Step Controllable Generator via Distillation-Friendly Single-Stream Teachers.**
Ngan Nguyen*, Dung Nguyen*, Quan Dao, Dimitris N. Metaxas, Minh Hoai, Cuong Pham, and Anh Tran.
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

3. **Cross-Space Distillation: Teaching One-Step Students with Modern Diffusion Teachers.**
Ngan Nguyen*, Anh Nguyen*, Duc Vu*, Trung Dao, Viet Nguyen, Quan Dao, Kien Nguyen, Chi Tran, Phong Nguyen, Khoi Nguyen, Cuong Pham, Dimitris N. Metaxas, Vishal M. Patel, and Anh Tran.
*European Conference on Computer Vision (**ECCV**)*, 2026.

2. **InverFill: One-Step Inversion for Enhanced Few-Step Diffusion Inpainting.**
Duc Vu*, **Ngan Nguyen***, Kien Nguyen*, Trong-Tung Nguyen*, Phong Nguyen, Khoi Nguyen, Cuong Pham, and Anh Tran.
*Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.

1. **Discrete Noise Inversion for Next-scale Autoregressive Text-based Image Editing (VARIN).**
Quan Dao, Xiaoxiao He, Ligong Han, **Ngan Nguyen**, Amin Heyrani Nobar, Faez Ahmed, Han Zhang, Viet Anh Nguyen, and Dimitris N. Metaxas.
*European Conference on Computer Vision (**ECCV**)*, 2026.

Preprints

2. **Beyond the VAE Bottleneck: One-Step Pixel Diffusion from Latent Teachers.**
Ngan Nguyen*, Duc Vu*, Cuong Pham, and Anh Tran.

1. **VideoDrift: Plug-and-Play Video Refinement for Diffusion Models via KV-Anchored Attention.**
Ngan Nguyen, Cuong Pham, and Anh Tran.

Research Experience

Qualcomm AI Research *July 2025 – Present*
AI Research Resident *Hanoi, Vietnam*

- **Efficient generative modeling:** Investigate one-step diffusion distillation, cross-space teacher–student transfer, controllable image generation, and inversion-based inpainting; co-authored work at NeurIPS, ECCV, and CVPR 2026. *Supervisors: Anh Tran and Cuong Pham.*
- **Pixel diffusion and video refinement:** Study VAE-free one-step pixel generation and plug-and-play video enhancement using inversion and KV-anchored attention (manuscripts under review).
- **Wearable AI memory (since August 2026):** Explore multimodal RAG, long-term memory, and retrieval for on-device/on-glass AI agents. *Supervisor: Dat Quoc Nguyen.*

References

Dr. Anh Tran
Principal Research Engineer, Qualcomm AI Research, Vietnam
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Assoc. Prof. Cuong Pham
Dean, Faculty of Artificial Intelligence; Director, PTIT.AI Research Lab, PTIT, Vietnam
Visiting Research Scientist, Qualcomm AI Research, Vietnam
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